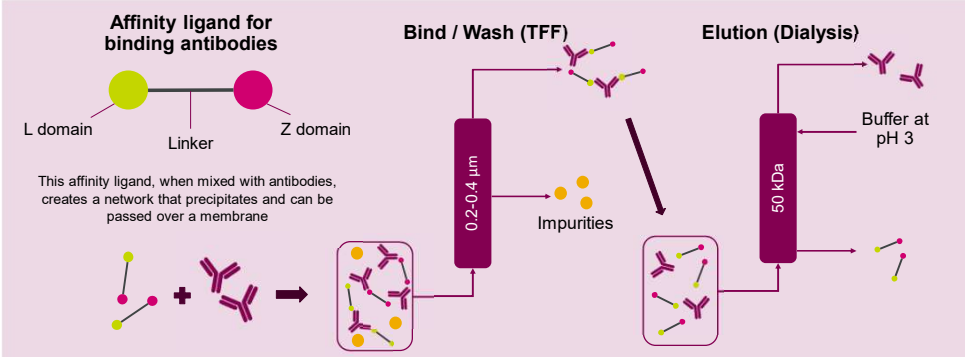


Designing and Optimizing Product Recovery in Continuous Counter-Current Dialysis (C3D) using Biophysical Methods

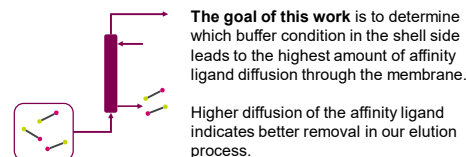
Taylor Bloom, Keith Morgan, Joel Graham, Ujwal Patil, Jon Coffman

AstraZeneca, Bioprocess Technologies and Engineering

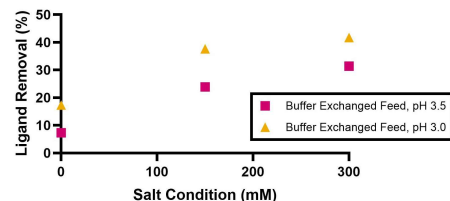
Introduction to Continuous Purification using Affinity Ligands



Evaluation of Buffer pH and Salt Condition



Results: Percent removal of the affinity ligand to the permeate at salt and pH conditions, using material from the bind / wash TFF step. Protein content was analyzed using a low pH SEC method.

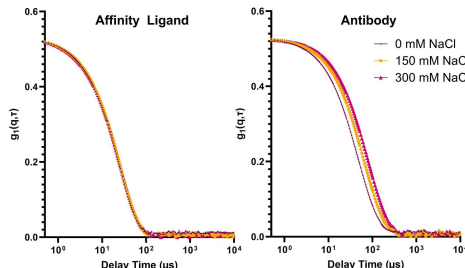


The best diffusion was achieved with a feed at 300 mM NaCl and pH 3.0. Therefore, this condition was selected for our process.

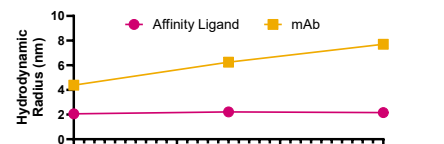
A lower pH buffer is assumed to support greater disassociation between the affinity ligand and the antibody.

Next, we performed dynamic light scattering (DLS) to understand why higher salt conditions lead to improved diffusion of the ligand.

Autocorrelation results: Normalized and manually fit autocorrelation data of the affinity ligand and monoclonal antibody in 100 mM Glycine, pH 3.0 buffer with three different salt conditions.

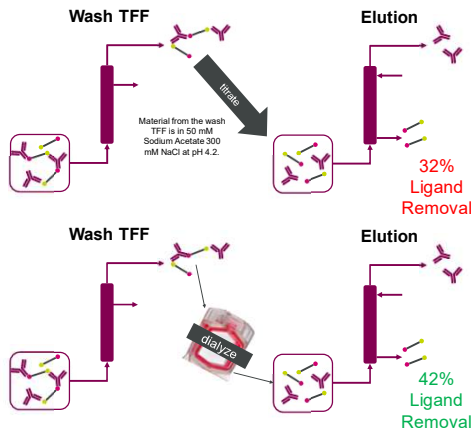


Radii results: Calculated radii of affinity ligand and monoclonal antibody in 100 mM Glycine, pH 3.0 buffer with three different salt conditions.

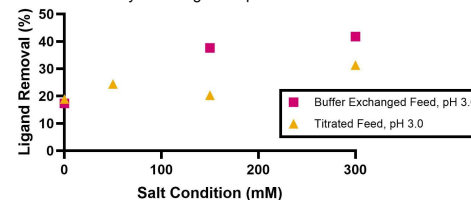


The salt-dependent increase in the antibody's apparent size likely reflects changes in conformational state and/or association behavior under the high ionic strength condition. This potentially reducing opportunities for nonspecific interactions between the two species, supporting the use of the higher salt buffer.

Titration vs. Buffer Exchanged Feed



Results: Percent removal of the affinity ligand to the permeate at various buffer conditions, using material from the bind / wash TFF step. Protein content was analyzed using a low pH SEC method.

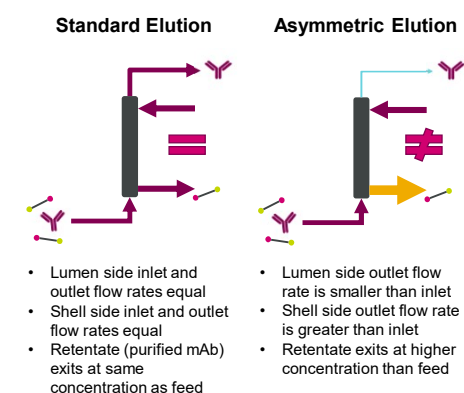


To get a better removal in the elution process, the material must be dialyzed into our elution buffer (100 mM Glycine 300 mM NaCl, pH 3.0) before the elution is run. Practically, this indicates that an intermediate membrane step is required to perform the buffer exchange before the elution step.

Conclusions

- 100 mM Glycine + 300 mM NaCl, pH 3.0 was chosen as the best shell-side buffer candidate, and this choice was validated by DLS
- An intermediate buffer exchange step is likely necessary to achieve desired ligand removal
- An asymmetric elution design allows for improved ligand removal

Asymmetric Dialysis for Elution



Results: Percent removal of the affinity ligand to the permeate using asymmetric elution design with two different feed mixtures

Feed	mAb:ligand ratio	Ligand Removal
Purified Antibody with Ligand Retentate	1:2	59%
Conditioned Media with Ligand Retentate	1:1	68%

The asymmetric elution design results in much higher ligand removal compared to the standard elution design. With two different feeds from the TFF outlet, we saw the highest removals so far. It is important to note that the change in mAb:ligand ratio of the material for the conditioned media run likely also boosted the removal.

References

- Coffman, J., Deldari, S., Patil, U., Ramos, I., Morgan, K., Bloom, T. (2023). Systems and methods for end to end continuous downstream processing. Patent No. WO202328097A1. World Intellectual Property Organization
- Patil, U., Chen, M., Ramos, I., et al. (2025). Asymmetric Dialysis: Truly unified single-pass ultrafiltration and buffer exchange. *Authorea*. <https://doi.org/10.22541/au.174910937.71524550/v1>
- Yeh, C. J., & Zydney, A. L. (2021). High performance countercurrent membrane purification for protein separations. *Journal of Membrane Science*, 633, 119396. <https://doi.org/10.1016/j.memsci.2021.119396>

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